20/07/2026

Practical 1

Aim: Study of various network commands used in Linux and windows:

Some important Linux networking commands

1. ip

The ip commands is one of the basic commands every administrator will need in daily work, from setting up new systems and assigning IPs to troubleshooting existing systems. The ip command can show address information, manipulate routing, plus display network various devices, interfaces and tunnels.

ip <OPTIONS><OBJECT><COMMAND>

Here are some common use cases for the ip command

a. To show the IP addresses assigned to an interface on your server:

[user123@localhost ~]$ ip address show

Output:

1: lo: <LOOPBACK, UP, LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever

2: enp0s3: <BROADCAST, MULTICAST, UP, LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:4e:ca:46 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3 valid_lft 84056sec preferred_lft 84056sec
    inet6 fe80::a00:27ff:fe4e:ca46/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

b. To assign an IP to an interface, for example.
enp0s3:
ip address add 192.168.1.254/24 dev enp0s3
Output:
RTNETLINK answers: Operation not permitted

c. To delete an IP on an interface
ip address del 192.168.1.254/24 dev enp0s3
Output:
RTNETLINK answers: Operation not permitted

d. Alter the status of the interface by bringing the interface eth0
ip link set eth0 up
Output:
Cannot find device "eth0"

e. Alter the status of the interface by bringing the interface eth0
ip link set eth0 down
Output:
Cannot find device "eth0"

f. Alter the status of the interface by ~~bringing the interface~~ enabling promiscuous mode for eth0
ip link set eth0 promise on
Output:
Error: either "dev" is duplicate, or "promise" is garbage

g. Add a default route (for all addresses) via the local 192.168.1.254 that can be reached on device eth0:
ip route add default via 192.168.1.254 dev eth0
Output:
Cannot find device "eth0"

h. Add a route to 192.168.1.0/24 via the gateway at 192.168.1.254:

ip route add 192.168.1.0/24 via 192.168.1.254

Output:

RTNETLINK answers: Operations not permitted

i. Add a route to 192.168.1.0/24 that can be reached on device eth0

ip route add 192.168.1.0/24 dev eth0

Output: Cannot find device eth0

j Delete the route for 192.168.1.0/24 via the gateway at 192.168.1.254:

ip route delete 192.168.1.0/24 via 192.168.1.254

Output:

RTNETLINK answers: Operations not permitted

k Display the route taken for IP 10.10.1.4:

ip route get 10.10.1.4

Output:

10.10.1.4 via 10.0.2.2 dev enp0s3 src 10.0.2.15

uid 1000

2. ifconfig

The ifconfig command is a staple in many sysadmin's tool belt for configuring and troubleshooting networks. It has since been replaced by the ip command discussed above.

Output:

enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500

inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255

inet6 fe80::a00:27ff:fe4e:ca46 prefixlen 64 scopeid 0x20<link>

ether 08:00:27:4e:ca:46 txqueuelen 1000 (Ethernet)

lo: flags =73<UP, LOOPBACK, RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scope id 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 18 bytes 2112 (2.0 KiB)
RX errors 0 dropped 0 overruns 0 frame 0

3. mtr

MTR (MattisTraceroute) is a network diagnostic tool that combines the features of ping and traceroute. It displays the route to a destination and provides statistics such as response time and packet loss for each hop, making it easier to identify network issues or faculty links.

a. The basic mtr command shows you the statistics, including each hop (hostnames) with time and loss%.

mtr google.com

Output:

My traceroute [V0.94]
localhost.localdomain (10.0.2.15) → google.com
2024-07-23T04:83:57+09

Keys: Help Display mode Restart statistics Order of fields quit

Packets Pings

Host Loss% Snt Last Avg Best Worst StDev

1. -gateway
2. (waiting for reply)
3. (waiting for reply)
4. (waiting for reply)
5. (waiting for reply)
6. (waiting for reply)
7. (waiting for reply)
8. (waiting for reply)
9. (waiting for reply)
10. (waiting for reply)
11. (waiting for reply)

12. (waiting for reply)
13. (waiting for reply)
~~14. (waiting for reply)~~
14. pnmaaa-ay-in-f14-1e100.net 0.0%. 563 15.0 8.8 7.2 32.6 2.4

b. Show numeric IP addresses (if you use -g, you can get IP addresses (numbers) instead of hostnames):

mtr -g google.com

Output:

mtr: invalid option -- 'g'

Usage:

mtr [options] hostname

-F, --filename FILE	read hostnames) from a file
-4	use IPV4 only
-6	use IPv6 only
-u, --udp	use UDP instead of ICMP echo
-T, --tcp	use TCP instead of ICMP echo
-I, --interface NAME	use named network interface
-h, --help	display this help and exit
-v, --version	output version information and exit

See the 'man 8 mtr' for details.

c. Show the numeric IP addresses and hostnames, too

mtr -b ~~gg~~ google.com

Output:

My traceroute [v0.94]

localhost.localdomain (10.0.2.15) → google.com

2026-07-23T04:50:28+05:30

Keys: Help Display mode Restart statistics Order of fields quit

Host	Loss%	Snt	Last	Avg	Best	Wrst	StDev
1. gateway (10.0.2.2)	0.0%	187	0.3	0.3	0.2	1.8	0.1

Packets Pings

2. (waiting for reply)
3. (waiting for reply)
4. (waiting for reply)
5. (waiting for reply)
6. (waiting for reply)
7. (waiting for reply)
8. (waiting for reply)
9. (waiting for reply)
10. (waiting for reply)
11. (waiting for reply)
12. (waiting for reply)
13 (waiting for reply)
14. pnmaa-az-in-f14.1e100.net (42.0.0%. 230 8-3 9.6 7.9
19.2 2

d. ~~tep dump~~ Set the number of pings that you want to send

mtr -c 10 google.com

Output:

My traceroute [V0.94]

localhost.localdomain (10.0.2.15) → google.com 2026-07-28T05:02:42+0530

Keys: Help Display mode Restart statistics Order of fields quit

Host	Packets Loss%	Snt	Last	Avg	Pings Best	Wrst	Stdev
1. _gateway	0.0%	4	0.5	0.4	0.3	0.5	0.1

2. (waiting for reply)
3. (waiting for reply)
4. (waiting for reply)
5. (waiting for reply)
6. (waiting for reply)
7. (waiting for reply)
8. (waiting for reply)

9. (waiting for reply)
10. (waiting for reply)
11. (waiting for reply)
12. (waiting for reply)
13. (waiting for reply)
14. pnmaaa-az.in-f14.1e100.net 0.0%. 3 9.2 8.9 8.7 9.2 0.3

4. tcpdump

The tcpdump command is designed for capturing and displaying packets.

If tcpdump is not installed :

```
dnf install -y tcpdump
```

Before starting any capture you need to know which interfaces tcpdump can use.

```
tcpdump -D
```

Output:

1. enp0s3 [Up, Running, Connected]
2. any (Pseudo-device that captures on all interfaces) [Up, Running]
3. lo [Up, Running, Loopback]
4. bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5. usbmon2 (Raw USB traffic, bus number 2)

If you want to capture traffic on eth0, you can initiate that with tcpdump -i enp0s3 sample

```
tcpdump -i enp0s3
```

Output:

tcp dump: enp0s3: You don't have permission to capture on that device.
(Socket: Operation not permitted)

tcpdump -i enp0s3 -c 10

Output:
tcpdump: enp0s3: You don't have permission to capture on that device (socket: Operation not permitted)

Capture traffic to and from one host
You can filter out traffic coming from a specific host. For example, to find traffic coming from and going to 8.8.8.8 use the command:

tcpdump -i ~~eth0~~ enp0s3 -c 10 host 8.8.8.8

Output
tcpdump: enp0s3: You don't have permission to capture on that device (socket: Operation not permitted)

For traffic coming from 8.8.8.8 use
tcp dump -i enp0s3 src host 8.8.88

Output:
tcpdump: enp0s3: You don't have permission to capture on that device (socket: Operation not permitted)

For outbound traffic going to 8.8.8.8, use
tcpdump -i ~~eth0~~ enp0s3 dst host 8.8.8.8

Output:
tcp dump: enp0s3: You dont have permission to capture on that device (socket: Operation not permitted)

Capture traffic to and from a network
You can also capture traffic to and from a specific network using the command below:

tcpdump -i enp0s3 net 10.1.0.0 mask 255.255.255.0

Output:
tcpdump: enp0s3: You don't have permission to capture on that device (socket: Operation not permitted)

Capture traffic to and from port numbers
Capture only DNS port 53 traffic:

tcpdump -i enp0s3 port 53

Output:
tcp dump: enp0s3: You don't have permission to capture on that device (socket: Operation not permitted)

For a specific host,

tcpdump -i enp0s3 host 8.8.8.8 and port 53

Output:

tcpdump: enp0s3: You don't have permission to capture on that device (socket: Operation not permitted)

To capture only HTTPS traffic,

tcpdump -i enp0s3 -c 10 host www.google.com and port 443

Output:

tcpdump: enp0s3: You dont have permission to capture on that device (socket: Operation not permitted)

To capture all port except port 80 and 25

tcpdump -i enp0s3 port not 53 and not 25

output:

tcpdump: enp0s3: You don't have permission to capture on that device (socket: Operation not permitted)

5. ~~Ping~~ ping

ping is a tool that verifies IP-level connectivity to another TCP/IP computer by sending Internet Control Message Protocol (ICMP) Echo Request messages. The receipt of corresponding Echo Reply messages is displayed, along with round trip times. ping is the primary TCP/IP command used to troubleshoot connectivity, reachability and name resolution.

ping google.com

Output:

PING google.com (142.250.193.14) 56(84) bytes of data.
64 bytes from tzdelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=1 ttl=111 time=41.6 ms
64 bytes from tzdelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=2 ttl=111 time=49.6 ms

If you want to ping a host ten times, use the following command:

ping -c 10 google.com

Output:

PING google.com (142.250.67.46) 56(84) bytes of data
64 bytes from pnmaaa-bb-in-f14.1e100.net (142.250.67.46):
icmp-seq=1 ttl=110 time=8.65ms
64 bytes from pnmaaa-bb-in-f14.1e100.net (142.250.67.46):
icmp-seq=2 ttl=110 time=8.17ms
64 bytes from pnmaaa-bb-in-f14.1e100.net (142.250.67.46):
icmp-seq=3 ttl=110 time=7.62ms

While pinging a host, ~~ten times~~ you'll find different output from the ping results, including the following three examples.

Destination Host Unreachable
The possible best reason is there is no route from the local host system and to the destination desired destination host, or a remote router reports that it has no route to the destination host

Request timed out
This result means that no Echo reply messages were received within the default time of one second or the time that you set while you are pinging that host. This can be due to many different causes, the most common include network congestion, failure of the ARP request, packet filtering/ firewall, etc.

Unknown host / Ping Request Could Not Find Host
Maybe you misspelled the hostname or the host does not exist at all in the network.

You must have 0% packet loss for every ping result with a good latency or lower response time. Depending on which transmission medium (UTP, fibre optics cable, Wi-Fi) you're using, your latency will differ.

Configuring an Ethernet connection by using nmcli

If you connect a host to the network over Ethernet, you can manage the connection's settings on the commandline by using nmcli utility.

Procedure

1. List the NetworkManager connection profiles
   nmcli connection show

   Output:

Name	UUID	TYPE	DEVICE
enpos3	b005d377-b846-4c9a-af16-e40d8a435187	ethernet	enpos3
lo	15b8ce4c-c2f8-4c73-9103-d8343216f730	loopback	lo

2. nmcli connection add con-name enpOS3 ifname enpos3 type ethernet

   Output:

   Not executed, as it creates a new network connection profile and modifies the existing network configuration

3. Optional: Rename the connection profile
   nmcli connection modify ~~an existing profile~~ "enpOS3"
   Here "enpOS3" is the name of the connection

   Not executed, as it modifies the existing network connection profile

4. Display the current settings of the connection profile
   nmcli connection show

   Output:

Name	UUID	TYPE	DEVICE
enpos3	b005d377-b846-4c9a-af16-e40d8a435187	ethernet	enpos3
lo	15b8ce4c-c2f8-4c73-9103-d8343216f730	loopback	lo

5. Configure the IPv4 settings:
   To use DHCP, enter:
   nmcli connection modify "enpOS3" ipv4.method auto
   Skip this step if ipv4.method is already set to auto (default)

   Not executed, as it changes the IPv4 configuration of the network connection.

To set a static IPV4 address, network mask, default gateway, DNS servers, and search domain enter:

nmcli connection modify "enp0s3" ipv4.method manual ipv4.addresses 192.0.2.1/24 ipv4.gateway 192.0.2.254 ipv4.dns 192.0.2.200 ipv4.dns-search example.com

Output:

Not executed, as it assigns a static IPv4 address, gateway, and DNS settings to the network connection

6. Configure the IPv6 settings:

To use stateless address auto configuration (SLAAC), enter:

nmcli connection modify "enp0s3" ipv6.method auto

Skip this step if ipv6.method is already set to auto (default)

Not executed as it changes the IPV6 configuration of the network connection

To set a static IPv6 address, network mask, default gateway, DNS servers, and search domain enter:

nmcli connection modify "enp0s3" ipv6.method manual ipv6.addresses 2001:db8:1::fffe/64 ipv6.gateway 2001:db8:1::fffe ipv6.dns 2000:db8:1::ffbb ipv6.dns-search example.com

Output:

Not executed as it assigns a static IPV6 address, gateway, and DNS settings to the network connection.

7. Activate the profile:

nmcli connection up Internal-LAN

Output:

Not executed as it activates the specified network connection profile and may alter the current network connection.

Verification:

1. Display the IP settings of the NIC:
   ip address show enpos●3

   Output:
   2: enpOS3: <BROADCAST, MULTICAST, UP, LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:4e:ca:46 brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 drd 10.0.2.255 scope global dynamic noprefixroute enpOS3 valid_lft 81616sec preferred lft 81616sec

2. Display the IPv4 default gateway
   ip route show ~~enp~~ default

   Output:
   default via 10.0.2.2 dev enpOS3 proto dhcp src 10.0.2.15 metric 100

3. Display the IPV6 default gateway:
   ip -6 route show default

   Output:
   NO OUTPUT

4. Display the DNS settings
   cat/etc/resolv.conf

   Output:
   # Generated by Network Manager
   nameserver 192.168.29.1

5. Use the ping utility to verify that this host can send packets to other hosts:
   Ping

   Output:
   PING google.com (142.250.193.4) 56(84) bytes of data.
   64 bytes from ~~tzdelb~~-bg-in-f14.1e100.net (142.250.193.14): icmp_seq

Troubleshooting

- Verify that the network cable is plugged in to the host and a switch.
- Check whether the link failure exists only on this host or also on other hosts connected to the same switch.
- Verify that the network cable and the network interface are working as expected. Perform hardware diagnosis steps and replace defect cables and network interface cards
- If the configuration on the disk does not match the configuration on the ~~def~~ device, stopping or restarting NetworkManager creates an in-memory connection that reflects the configuration of the device.

Basic Networking Commands

1. arp -a

ARP is a short form of address resolution protocol, It will show the IP address of your computer along with the IP address and MAC address of your router.

Output:

Interface: 172.16.11.1 ---- 0x10

Internet Address	Physical Address	Type
172.16.8.1	7c-5a-1c-cf-be-45	dynamic
172.16.8.32	d4-3d-7e-ad-41-67	dynamic
172.16.8.38	d4-3d-7e-ad-39-52	dynamic
172.16.8.52	f8-bc-12-8d-e4-2b	dynamic

Interface: 192.168.56.1 ---- 0x11

Internet Address	Physical Address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static

2. hostname

This is the simplest of all TCP/IP commands. It simply displays the name of your computer.

Output:

LAPTOP-660VCTK4

3. ipconfig/all

This command displays detailed configuration information about your TCP/IP connection including Router, Gateway, DNS, DHCP and type of Ethernet adapter in your system

Output:

```
Windows IP Configuration

Host Name . . . . . : LAPTOP-660VCTK4
Primary Dns Suffix . . . . . :
Node Type . . . . . : Mixed
IP Routing Enabled . . . . . : No
WINS Proxy Enabled . . . . : No
```

4. nbtstat -a

This command helps solve problems with NetBIOS name resolution. (Nbt stands for NetBION over TCP/IP)

Output:

Displays protocol statistics and current TCP/IP connections using NBT (NetBIOS over TCP/IP).

NBTSTAT [[-a Remote Name] [-A IP address] [-c] [-n] [-r] [-R] [-RR] [-s] [-S] [interval]]

-a (adapter status) Lists the remote machine's name table given its name

-A (Adapter Status) Lists the remote machine's name table given its IP address

-c (cache) Lists NBT's cache of remote [machine] names and their IP addresses

-n (names) Lists Local NetBIOS names

5. netstat: (network statistics) netstat displays a variety of statistics about a computers active TCP/IP connections It is a command line tool for monitoring network connections both incoming and outgoing as well as viewing routing tables, interface statistics etc. eg: netstat -r

Output:

Active Connections			
Proto	Local Address	Foreign Address	State
TCP	172.16.11.1:49443	4.213.25.241:https	ESTABLISHED

6. nslookup: (name server lookup) is a tool used to perform DNS lookups in Linux. It is used to display DNS details, such as the IP address of a particular computer, the MX records for a domain or the NS servers of a domain. nslookup can operate in two modes: interactive and non-interactive
eg: nslookup www.google.com

Output:
Default Server: Unknown
Address: 172.16.8.1

7. pathping
Pathping is unique to Window's and is basically a combinations of the ping and Tracert commands
Pathping traces the route to the destination address then launches a 25 second test of each router along the way, gathering statistics on the rate of data lossalong each hop.

Output:
Usage: pathping [-g host-list][-h maximum_hops][-i address] [-n]

[-p period] [-q num_queries] [-w timeout]
[-4] [-6] target_name.

Options:
-g host-list Loose source route along host-list
-h maximum_hops Maximum number of hops to search for target

8. ping
(Packet INternet Groper) command is the best way to test connectivity between two nodes. Ping uses ICMP (Internet Control Message Protocol) to communicate to other devices

1. ping hostname (ping local host)
2. ping ip address (ping 4.2.2.2)
3. ping fully qualified domain name (ping www.facebook.com)

Output: ping LAPTOP-660VCTK4

Pinging LAPTOP-660VCTK4 [fe80::e07e:6cfa:ea8b:c89e%16] with ~~timeouts~~
32 bytes of data:

Reply from fe80::e07e:6cfa:ea8b:c89e%16
time<1ms

Reply from fe80::e07e:6c7e:6cfa:ea8b:
c89e%16: time<1ms

Reply from fe80::e07e:6cfa:ea8b:c89e%16:
time<1ms

Reply from fe80::e07e:6cfa:ea8b:c89e%16:
time<1ms

9. Route
route command is used to show/manipulate the IP routing table. It is primarily used to setup static routes to specific host or networks via an interface.

Output:

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
[MASK netmask] [gateway] [METRIC metric]
[IF interface]

-f Clears the routing tables of all gateway entries. If this is used in conjunction with one of the commands the table are cleared prior to running the command.

-p When used with the ADD command, makes a route persistent across boots of the system. By default, routes are not preserved when the system is restarted. Ignored for all other commands, which always affect the appropriate persistent routes.

-4 Force using IPv4

-6 Force using IPv6

Result:

Thus the various network commands in Linux & Windows are studied and executed successfully.